

PRAKTIKUM SISTEM OPERASI

MODUL 10

shell



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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JURNAL

1. Soal 1

[40 Poin] Akan dimodifikasi shell dengan modifikasi syscall bernama **chname** yang berfungsi untuk mengubah nama suatu proses. Lihat kembali modul sebelumnya cara membuat syscall.

Perhatikan sekarang syscall **chname** mempunyai 3 parameter yaitu pid, character dan panjang character. Character untuk menyimpan nama dan panjang character untuk panjang nama.

- a. Pada prototypes.h chname diubah menjadi:

```
/* in file chname.c */
extern pril6 chname(pid32, char *, uint32);
```

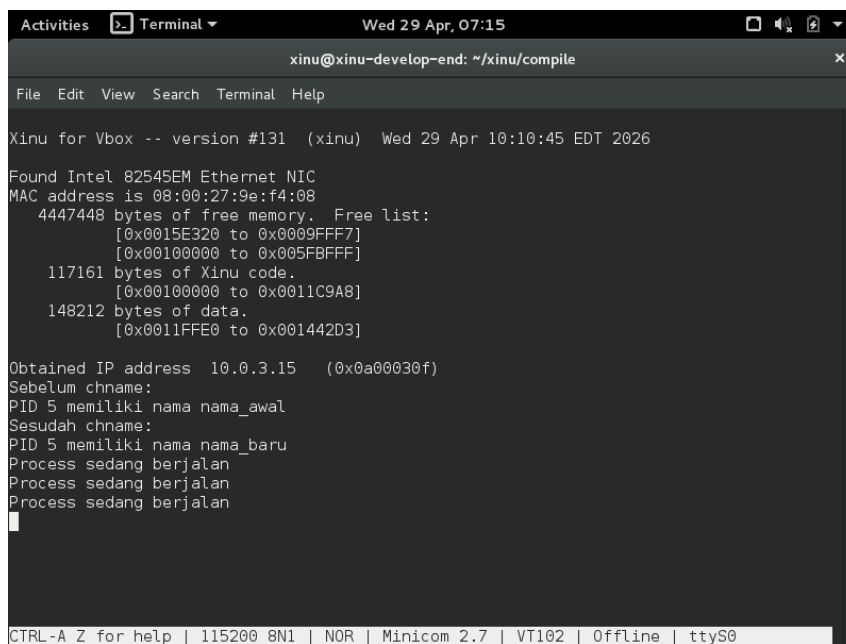
- b. Pada chname.c fungsi diubah dari:

```
pril6 chname(
    pid32      pid,      /* ID of process to change */
    pril6      newprio   /* New priority */
)
```

menjadi

```
pril6 chname(
    pid32      pid,      /* ID of process to change */
    char       *newname,
    uint32      namelen
)
```

Modifikasi kode pada chname.c sehingga nama proses bisa diubah bila syscall tersebut dipanggil.



```
Activities Terminal Wed 29 Apr, 07:15
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

Xinu for Vbox -- version #131 (xinu) Wed 29 Apr 10:10:45 EDT 2026

Found Intel 82545EM Ethernet NIC
MAC address is 08:00:27:9e:f4:08
4447448 bytes of free memory. Free list:
  [0x0015E320 to 0x0009FFFF]
  [0x00100000 to 0x005FBFFF]
117161 bytes of Xinu code.
  [0x00100000 to 0x0011C9A8]
148212 bytes of data.
  [0x001FFE0 to 0x001442D3]

Obtained IP address 10.0.3.15 (0x0a00030f)
Sebelum chname:
PID 5 memiliki nama nama_awal
Setelah chname:
PID 5 memiliki nama nama_baru
Process sedang berjalan
Process sedang berjalan
Process sedang berjalan

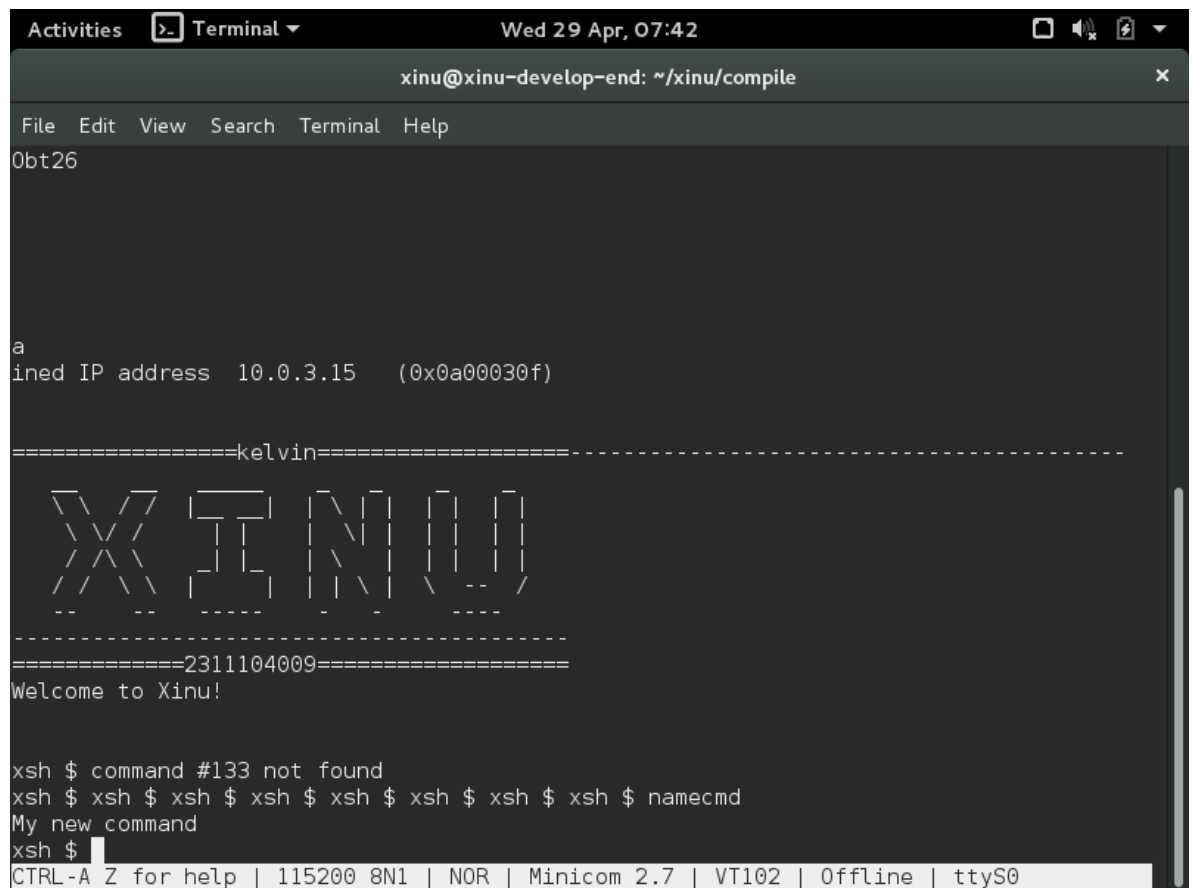
```

2. Soal 2

2. [40 Poin] Buatlah perintah baru bernama namecmd sesuai dengan langkah-langkah pada no.5 pada modul shell!

Berikut adalah kode dalam perintah baru namecmd:

```
printf("My new command");  
  
// pid 3 = netin; mengubah nama proses pid 3 yaitu netin menjadi hello  
chname(3,"hello",5);  
  
return 0;
```



The screenshot shows a terminal window titled "Terminal" with the date and time "Wed 29 Apr, 07:42". The user is logged in as "xinu" on a machine named "xinu-develop-end" at the directory "~/xinu/compile". The terminal output shows the following sequence of events:

- The user enters "0bt26".
- The user enters "a".
- The terminal displays "ined IP address 10.0.3.15 (0x0a00030f)".
- The terminal displays a separator line "=====kelvin=====".
- The terminal displays the ASCII art "XINU".
- The terminal displays another separator line "=====2311104009=====".
- The terminal displays "Welcome to Xinu!".
- The user enters "xsh \$ command #133 not found".
- The user enters "xsh \$ xsh \$ xsh \$ xsh \$ xsh \$ xsh \$ xsh \$ namecmd".
- The terminal displays "My new command".
- The user enters "xsh \$".

The terminal status bar at the bottom shows "CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0".

3. Soal nomor 3

3. [20 Poin] Test hasilnya:

- Masuk ke terminal xinu
- Jalankan perintah ps
- Jalankan perintah namecmd
- Jalankan perintah ps
- Lihat nama proses telah berubah

```
Activities Terminal Wed 29 Apr, 07:51
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help
=====2311104009=====
Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout           wait  500   0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500   0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process    recv  20    3 0x005E7FFC 0x005E7F44 65536 0 0
5 shell           recv  50    4 0x005D7FFC 0x005D79AC 8192 0 0
6 ps              curr  20    5 0x005F7FFC 0x005F7DD8 8192 0 0

xsh $ namecmd
My new command
xsh $ pas
command pas not found
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout           wait  500   0 0x005FBFFC 0x005FBF30 8192 0 0
2 helo            wait  500   0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process    recv  20    3 0x005E7FFC 0x005E7F44 65536 0 0
5 shell           recv  50    4 0x005D7FFC 0x005D79AC 8192 0 7
8 ps              curr  20    5 0x005F7FFC 0x005F7FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```